

Q.1 $a = a + 1$; can be written as _____.

- A. $a + 1$;
- B. $a + a + 1$;
- C. $a += 1$;**
- D. $1 + a = a$;

Q.2 Why is it a good practice to write macro names in capital letters?

- A. It helps distinguish macros from functions and variables**
- B. It speeds up macro execution
- C. It is required by the compiler
- D. It avoids runtime errors

Q.3 Which of the following code segment represents a left shift operator?

- A. `cout<<1;`
- B. `result = number >> 1 ;`
- C. `cin >> number ;`
- D. `result = number << 1 ;`**

Q.4 Default constructor takes _____.

- A. one parameter
- B. character type parameter
- C. no parameters**
- D. two parameters

Q.5 the sizeof operator is used to determine the size of _____.

- A. None of the given
- B. data type**
- C. variable
- D. data

Q.6 Destructor _____.

- A. takes no arguments and can be overloaded
- B. cannot be overloaded and have return type
- C. cannot be overloaded and have no return type**
- D. can be overloaded and have return type

Q.7 A class is _____.

A. a user defined data type

B. a member function

C. a built in function

D. an array

Q.8 For accessing data members we use _____ operator

A. plus +

B. dot .

C. multiplication *

D. Division /

Q.9 What is the main difference between a struct and a class in C++?

A. Structs can't have functions.

B. Classes don't allow data grouping.

C. There is no difference.

D. Struct members are public by default; class members are private.

Q.10 Which is not true regarding structures?

A. a) We can declare arrays of structure

B. b) We can take the size of structure

C. c) We can't declare pointers to structure

D. d) Both b & c

Q.1 Which of the following is the default mode of the file opening while using the ofstream class?

- A. ios::out**
 - B. ios::in
 - C. ios::app
 - D. ios::trunc
-

Q.2 Default constructor generated by _____ does _____ for us.

- A. compiler , initialization
 - B. classes , initialization
 - C. main program, memory management
 - D. compiler, no initialization**
-

Q.3 student s[100]; means that 's' is an array of hundred students of type 'student' structure. What will be the correct statement to access the 'name' data element of the fifty sixth student from the array?

- A. s[55].name;**
 - B. s.name[56];
 - C. s[56].name;
 - D. s.name[55];
-

Q.4 What is the storage space of double data type?

- A. 64 bytes
 - B. 16 bytes
 - C. 8 bytes**
 - D. 32 bytes
-

Q.5 a = a + 1; can be written as _____.

- A. 1 + a = a;
 - B. a + 1;
 - C. a += 1;**
 - D. a + a + 1;
-

Q.6 Public member functions of the class _____ main program.

- A. are accessible in**
 - B. are defined within the
 - C. are not accessible in
 - D. are private to
-

Q.7 `a = a & b;` can be written as_____.

A. `a &= b;`

B. `a = b;`

C. `a =& b;`

D. `a=a & b;`

Q.8 _____ is a special type of pointer we have to cast it before we use it.

A. float

B. double

C. void

D. integer

Q.9 For casting, we normally declare a pointer of type _____.

A. NULL

B. we are going to use

C. void

D. integer

Q.10 The main advantage of function overloading is that _____.

A. The function becomes inline

B. The program becomes portable

C. The program becomes more readable

D. The program becomes complex

Q.1 _____ function gives the position of the next character to be read from that file.

- A. seekp()
 - B. seekg()
 - C. tellg()**
 - D. tellp()
-

Q.2 A pointer with value NULL is defined in the header files _____ and _____.

- A. conio.h , iostream.h
 - B. stdlib.h , stddef.h**
 - C. process.h, stdi.h
 - D. conio.c , getche()
-

Q.3 Destructor _____.

- A. takes no arguments and can be overloaded
 - B. cannot be overloaded and have no return type**
 - C. cannot be overloaded and have return type
 - D. can be overloaded and have return type
-

Q.4 Macros are categorized into _____ type(s).

- A. two**
 - B. one
 - C. three
 - D. four
-

Q.5 What does the calloc function return if it fails to allocate memory?

- A. NULL**
 - B. It crashes the program
 - C. 0
 - D. -1
-

Q.6 We use dot operator to access the data members of both _____ and _____.

- A. unions, arrays
 - B. unions, structures**
 - C. arrays, structures
 - D. none of them
-

Q.7 Which of the following permits function overloading in C++.

- A. Number of arguments only
 - B. Both type and arguments**
 - C. Type only
 - D. Number of function names
-

Q.8 Which is the correct syntax to define a structure named 'student'?

- A. structure student(
//variables
 - B. struct student{
//variables**
 - C. structure student{
//variables
 - D. struct student(
//variables
-

Q.9 Object code is machine code but it is not _____ and _____.

- A. tested , compiled
 - B. faster , efficient
 - C. relocatable , executable**
 - D. compiled ,debugged
-

Q.10 The dynamic memory allocation uses _____ whereas static memory allocation uses _____.

- A. classes , array
 - B. heap , stack**
 - C. array , stack
 - D. stack , lists
-

Q.1 Look at the program code and identify the error. #include<iostream> using namespace std; #define PI 3.1415926; main() { int radius = 5; cout << "Area of circle with radius " << radius << " = " << PI * radius * radius; }

- A. The value of PI cannot be defined in the beginning
- B. It will be compiled and returns nothing
- C. It will be compiled successfully and will not generate error
- D. Error exists in line number 2. Semi colon is not allowed with define directive**

Q.2 _____ is the pointer which determines the position in a file from where the next read operation occurs.

- A. Tell
- B. Set
- C. Put
- D. Get**

Q.3 Default constructor generated by _____ does _____ for us.

- A. compiler , initialization
- B. classes , initialization
- C. compiler, no initialization**
- D. main program, memory management

Q.4 By writing aFile.seekg(15L, ios::cur) we are moving _____.

- A. 15 bytes in the backward direction starting from the current position
- B. 15 bytes in the forward direction starting from the current position**
- C. 15 bits in the backward direction starting from the previous position
- D. 15 bits in the backward direction starting from the current position

Q.5 Destructor _____.

- A. can be overloaded and have return type
- B. cannot be overloaded and have return type
- C. cannot be overloaded and have no return type**
- D. takes no arguments and can be overloaded

Q.6 We have opened a file stream myfile for reading (getting), myfile.tellg () gives us the current get position of the file pointer. It returns a whole number of type_____

- A. short
- B. double
- C. int
- D. long**

Q.7 Sometimes after allocating memory we need additional space, for this purpose we use _____.

- A. reallocation function
 - B. void pointer
 - C. realloc function**
 - D. calloc function
-

Q.8 How many bits form a byte?

- A. 2
 - B. 6
 - C. 8**
 - D. 4
-

Q.9 In a class we can have _____ constructor(s).

- A. many**
 - B. two
 - C. only one
 - D. no
-

Q.10 What is the default access level for members of a class in C++?

- A. public
 - B. protected
 - C. global
 - D. private**
-

Q.1 _____ allow us to have a memory location and use it as int or char interchangeably

- A. union
- B. None of the given
- C. structures
- D. construct

Q.2 Which one of the following is used to perform bit-wise exclusive OR operation?

- A. ~
- B. ||
- C. |
- D. ^

Q.3 A file opened with the help of _____ can be used for both input and output.

- A. fstream
- B. ofstream
- C. ifstream
- D. iostream

Q.4 What is the main advantage of using macros instead of functions for simple tasks?

- A. Macros compile faster
- B. Macros reduce memory usage
- C. Macros avoid function call overhead
- D. Macros are more secure

Q.5 _____ for parameters is also done for inline functions.

- A. Automatic type checking
- B. Syntax checking
- C. Value checking
- D. Testing

Q.6 Which one of the following is mandatory preprocessor directive for c++?

- A. #undef <iostream>
- B. #include <iostream>
- C. All of the given
- D. #undef <iostream>

Q.7 Initializing the data member within the class is _____.

- A. a logical error
 - B. a run time error
 - C. not an error**
 - D. a syntax error
-

Q.8 Which of the following memory allocation types occurs at compile-time?

- A. Heap Allocation
 - B. Static Memory Allocation**
 - C. Runtime Allocation
 - D. Dynamic Memory Allocation
-

Q.9 What can the resultant number when you AND two numbers 12 & 8 bit-wise using & operator?

- A. 4
 - B. 6
 - C. 12
 - D. 8**
-

Q.10 _____ is a special type of pointer we have to cast it before we use it.

- A. integer
 - B. double
 - C. void**
 - D. float
-

Q.1 When accessing a structure member, the identifier to the left of the dot operator is the name of _____.

- A. Structure variable
- B. Structure tag
- C. Structure member
- D. Structure data

Q.2 Destructor _____.

- A. cannot be overloaded and have return type
- B. can be overloaded and have return type
- C. cannot be overloaded and have no return type
- D. takes no arguments and can be overloaded

Q.3 Name the function whose definition can be substituted at a place where its function call is made _____.

- A. friend function
- B. static function
- C. volatile function
- D. inline function

Q.4 What term describes creating variables from a class?

- A. Object creation
- B. Declaration
- C. Initialization
- D. Implementation

Q.5 What does the calloc function return if it fails to allocate memory?

- A. It crashes the program
- B. -1
- C. NULL
- D. 0

Q.6 Bugs can occur due to _____

- A. initialization of a variable
- B. uninitialized data
- C. object destruction
- D. default values

Q.7 Which of the following code segment represents a left shift operator?

- A. `result = number >> 1 ;`
 - B. `cout<<1;`
 - C. `cin >> number ;`
 - D. `result = number << 1 ;`**
-

Q.8 Which of the following permits function overloading in C++.

- A. Number of arguments only
 - B. Number of function names
 - C. Type only
 - D. Both type and arguments**
-

Q.9 student s[100]; means that 's' is an array of hundred students of type 'student' structure. What will be the correct statement to access the 'name' data element of the fifty sixth student from the array?

- A. `s[55].name;`**
 - B. `s[56].name;`
 - C. `s.name[55];`
 - D. `s.name[56];`
-

Q.10 A file opened with the help of _____ can be used for both input and output.

- A. `iostream`
 - B. `ofstream`
 - C. `fstream`**
 - D. `ifstream`
-

Q.1 Public member functions of the class _____ main program.

- A. are defined within the
- B. are private to
- C. are accessible in**
- D. are not accessible in

Q.2 malloc function returns _____ to the _____ from the available memory.

- A. void pointer, starting of the chunk of memory allocated from the array
- B. integer pointer, middle of the chunk of memory allocated from the heap
- C. void pointer , starting of the chunk of memory allocated from the heap**
- D. NULL pointer, end of the chunk of memory allocated from the stack

Q.3 The function overloading requires _____.

- A. the argument list to be the same
- B. the argument list to be different.**
- C. the arguments of integer type only
- D. no arguments

Q.4 For converting a void type pointer ptr into integer type, the correct syntax is:

- A. int * ptr
- B. *int ptr
- C. ptr * int
- D. (int *) ptr**

Q.5 When accessing a structure member, the identifier to the left of the dot operator is the name of _____.

- A. Structure data
- B. Structure variable**
- C. Structure member
- D. Structure tag

Q.6 Multiplicatiive, Additive and assignment operators are ____ operators.

- A. Conditional
- B. Relational
- C. Binary**
- D. Unary

Q.7 Name the function whose definition can be substituted at a place where its function call is made _____.

- A. static function
 - B. inline function**
 - C. volatile function
 - D. friend function
-

Q.8 What does the calloc function return if it fails to allocate memory?

- A. It crashes the program
 - B. NULL**
 - C. -1
 - D. 0
-

Q.9 _____ for parameters is also done for inline functions.

- A. Value checking
 - B. Automatic type checking**
 - C. Testing
 - D. Syntax checking
-

Q.10 Initializing the data member within the class is _____.

- A. a logical error
 - B. not an error**
 - C. a syntax error
 - D. a run time error
-

Q.1 When accessing a structure member, the identifier to the left of the dot operator is the name of _____.

- A. Structure member
- B. Structure tag
- C. Structure data
- D. Structure variable**

Q.2 What is the main difference between a struct and a class in C++?

- A. Structs can't have functions.
- B. There is no difference.
- C. Classes don't allow data grouping.
- D. Struct members are public by default; class members are private.**

Q.3 These functions seekg() and seekp() requires an argument of type _____ to determine that how many bytes to be move forward or backward.

- A. int
- B. long**
- C. short
- D. double

Q.4 _____ for parameters is also done for inline functions.

- A. Syntax checking
- B. Automatic type checking**
- C. Testing
- D. Value checking

Q.5 The object code of our program is combined with the _____.

- A. source program
- B. with header files
- C. machine code of the operating system
- D. object code of the library functions**

Q.6 A structure is a collection of _____ under a single name.

- A. values
- B. variables
- C. data**
- D. None of the given

Q.7 The malloc function takes _____ argument(s).

- A. four
 - B. one**
 - C. two
 - D. three
-

Q.8 What does the calloc function return if it fails to allocate memory?

- A. NULL**
 - B. 0
 - C. It crashes the program
 - D. -1
-

Q.9 The exclusive OR operator returns 1 if _____.

- A. Both inputs are 0
 - B. None of the given
 - C. Both inputs are 1
 - D. One input is 1 and other is zero**
-

Q.10 The memory allocation functions return a chunk of memory with a pointer of type_____.

- A. integer
 - B. float
 - C. void**
 - D. ptr
-

Q.1 Which header file is required to use file I/O operations?

- A. <fstream>**
 - B. <iostream>
 - C. <ifstream>
 - D. <ostream>
-

Q.2 What can the resultant number when you AND two numbers 12 & 8 bit-wise using & operator?

- A. 8**
 - B. 4
 - C. 6
 - D. 12
-

Q.3 Once we have defined a symbolic constant value using #define, that value _____ during program execution.

- A. becomes zero
 - B. can be changed
 - C. varies
 - D. cannot be changed**
-

Q.4 What type of memory is used when you allocate memory dynamically in C?

- A. Stack
 - B. Heap**
 - C. Data Segment
 - D. Code Segment
-

Q.5 For casting, we normally declare a pointer of type _____.

- A. NULL
 - B. void
 - C. we are going to use**
 - D. integer
-

Q.6 Which one is the correct example of a unary operator?

- A. -**
 - B. /
 - C. ==
 - D. &
-

Q.7 In shifting operations if zero is inserted at the left most bit, the negative number will become a ____ number.

- A. Greater
 - B. Positive**
 - C. Smaller
 - D. Negative
-

Q.8 Macros are categorized into _____ type(s).

- A. three
 - B. two**
 - C. four
 - D. one
-

Q.9 What is the default access level for members of a class in C++?

- A. protected
 - B. public
 - C. private**
 - D. global
-

Q.10 Object code is machine code but it is not _____ and _____.

- A. compiled ,debugged
 - B. tested , compiled
 - C. relocatable , executable**
 - D. faster , efficient
-

Q.1 Which one is the correct example of a unary operator?

- A. &
 - B. -**
 - C. ==
 - D. /
-

Q.2 For accessing data members we use _____ operator

- A. Division /
 - B. dot .**
 - C. multiplication *
 - D. plus +
-

Q.3 A file opened with _____ is used for input and cannot be used for output.

- A. conio
 - B. iostream
 - C. ifstream**
 - D. ofstream
-

Q.4 In _____, we have different data members and all of these have their own memory space.

- A. union
 - B. None of the given
 - C. structures**
 - D. construct
-

Q.5 What is encapsulation in the context of classes?

- A. Making everything public
 - B. Grouping unrelated data
 - C. Grouping functions only**
 - D. Hiding data implementation details
-

Q.6 Which of the following code segment represents a left shift operator?

- A. result = number << 1 ;**
 - B. cin >> number ;
 - C. cout<<1;
 - D. result = number >> 1 ;
-

Q.7 By writing `aFile.seekg(15L, ios::cur)` we are moving _____.

- A. 15 bits in the backward direction starting from the current position
 - B. 15 bytes in the forward direction starting from the current position**
 - C. 15 bytes in the backward direction starting from the current position
 - D. 15 bits in the backward direction starting from the previous position
-

Q.8 The function overloading requires _____.

- A. the arguments of integer type only
 - B. the argument list to be different.**
 - C. no arguments
 - D. the argument list to be the same
-

Q.9 The heap memory structure _____.

- A. constantly changes in size**
 - B. remains constant
 - C. is just like stack
 - D. contains no space
-

Q.10 Which one of the following is mandatory preprocessor directive for c++?

- A. All of the given
 - B. `#undef <iostream>`
 - C. `#undef <iostream>`
 - D. `#include <iostream>`**
-

Q.1 How is #include "filename" different from #include <filename>?

- A. It includes only header files with .cpp extension
 - B. It uses a different syntax with no functional difference
 - C. It includes binary files only
 - D. It searches the current directory first**
-

Q.2 _____ is the pointer which determines the position in a file from where the next read operation occurs.

- A. Get
 - B. Tell**
 - C. Put
 - D. Set
-

Q.3 When accessing a structure member, the identifier to the left of the dot operator is the name of _____.

- A. Structure member
 - B. Structure tag
 - C. Structure data
 - D. Structure variable**
-

Q.4 Structures are syntactically defined with the word_____.

- A. structure
 - B. struc
 - C. None of the given
 - D. struct**
-

Q.5 _____function is used to determine the next position to write a character while writing into a file.

- A. seekg()
 - B. seekp()**
 - C. tellp()
 - D. tellg()
-

Q.6 The compiler use a name mangling technique to _____.

- A. Run the functions
 - B. Compile the functions
 - C. Generate a unique token that is assigned to each function**
 - D. Generate unique functions
-

Q.7 $a = a + 1$; can be written as _____.

- A. $a + a + 1$;
- B. $a + 1$;**
- C. $1 + a = a$;
- D. $a + = 1$;

Q.8 In shifting operations if zero is inserted at the left most bit, the negative number will become a _____ number.

- A. Negative
- B. Greater
- C. Smaller
- D. Positive**

Q.9 A pointer with value NULL is defined in the header files _____ and _____.

- A. conio.h , iostream.h
- B. process.h, stdi.h
- C. conio.c , getche()
- D. stdlib.h , stddef.h**

Q.10 The function calloc takes two arguments, first argument is the _____ and the second argument is the _____.

- A. space in terms of string character, size of list
- B. size of array , number of bytes
- C. required space in terms of numbers , size of space**
- D. size of space, number of bytes

Q.1 The function calloc takes two arguments, first argument is the _____ and the second argument is the _____.

- A. space in terms of string character, size of list
- B. size of space, number of bytes
- C. required space in terms of numbers , size of space
- D. size of array , number of bytes**

Q.2 Which of the following memory allocation types occurs at compile-time?

- A. Heap Allocation
- B. Dynamic Memory Allocation
- C. Static Memory Allocation**
- D. Runtime Allocation

Q.3 _____ is the pointer which determines the position in a file from where the next read operation occurs.

- A. Tell**
- B. Get
- C. Set
- D. Put

Q.4 To access the data members of structure, _____ is used.

- A. Logical operator
- B. Dot operator**
- C. Dereference operator
- D. Address operator

Q.5 What is the main purpose of the C preprocessor?

- A. Enhance the C language by preprocessing code before compilation**
- B. Translate object code into machine code
- C. Optimize the executable file
- D. Execute the program

Q.6 The data members of the class are initialized _____.

- A. within main program
- B. at compile time
- C. outside the function
- D. at runtime**

Q.7 Constructor is itself a _____ of C++ and _____.

- A. object, can not be initialized
 - B. function , cannot be overloaded
 - C. function, can be overloaded**
 - D. class , can be overloaded
-

Q.8 By writing aFile.seekg(15L, ios::cur) we are moving _____.

- A. 15 bits in the backward direction starting from the previous position
 - B. 15 bytes in the forward direction starting from the current position**
 - C. 15 bits in the backward direction starting from the current position
 - D. 15 bytes in the backward direction starting from the current position
-

Q.9 Constructor is special type of function :

- A. which returns NULL pointer
 - B. which has no return type**
 - C. which returns integer type data
 - D. which returns zero
-

Q.10 What is the storage space of double data type?

- A. 16 bytes
 - B. 32 bytes
 - C. 64 bytes
 - D. 8 bytes**
-

Q.1 What is the sequence of event(s) when allocating memory to an object using new operator?

- A. Only constructor is called for object
- B. Only block of memory is allocated for object
- C. Memory is allocated first before calling constructor**
- D. Constructor is called first before allocating memory

Q.2 date+=1; is equivalent to _____.

- A. ++date
- B. date+1
- C. date++**
- D. 1+date

Q.3 Consider the following code segment. Which of the following will be called while executing code at line 2?

```
String s1, s2;  
s1 = s2;  
s1 = s2;
```

- A. Assignment operator
- B. Parameterized constructor
- C. Copy constructor**
- D. Default constructor

Q.4 Which statement is true about the following code in C++? int &j = i;

- A. The value of variable i is assigned to the variable j
- B. The address of variable j is assigned to the variable i
- C. The address of variable i is assigned to the variable j
- D. j is a synonym of variable i**

Q.5 A class can be declared as a _____ of other class.

- A. member
- B. member function
- C. friend**
- D. part

Q.6 In the scenario, if we want to grant access to the private data members of a Class1 to the Class2, then which statement is true about this scenario?

- A. Class2 is required to be friend of Class1**
- B. Class2 inherits from Class1
- C. Class1 inherits from Class2
- D. Class1 is required to be friend of Class2

Q.7 The memory allocation in C++ is carried out with the help of _____.

- A. NULL pointer
 - B. new operator**
 - C. + operator
 - D. dot operator
-

Q.8 Which of the following is the correct definition of streams?

- A. It is an ordered sequence of bytes**
 - B. It is an unordered sequence of bytes
 - C. It is an unordered sequence of bits
 - D. It is an ordered sequence of bits
-

Q.9 When the compiler overloads the assignment (=) operator by default then

- A. Compiler does not do member wise assignment.
 - B. Compiler does not allow default assignment operator
 - C. Compiler does member wise assignment.**
 - D. Class members are not assigned properly
-

Q.10 new operator can be used for _____.

- A. only integer data type
 - B. dot operator
 - C. only char and integer data types
 - D. integer , float, char and double data types**
-

Q.1 What is the difference between cout and cerr?

- A. cout is unbuffered output and cerr is buffered output
 - B. cout is standard output and cerr is not a standard output
 - C. cout is not a standard output and cerr is standard output
 - D. cout is buffered output and cerr is unbuffered output**
-

Q.2 Functions declared with the _____ specifier in a class member list are called of that class.

friend functions

- A. public
 - B. friend**
 - C. protected
 - D. private
-

Q.3 *this is a pointer that always points to

- A. Current pointer of the class
 - B. A Value of data member
 - C. Current function of the class
 - D. Current object of the class**
-

Q.4 _____ will return the number of bytes reserved for a variable or data type.

- A. sizeof operator**
 - B. free operator
 - C. new operator
 - D. void pointer
-

Q.5 new operator can be used for _____.

- A. integer , float, char and double data types**
 - B. only char and integer data types
 - C. only integer data type
 - D. dot operator
-

Q.6 The concept of _____ allows us to separate the interface from the implementation of the class.

- A. encapsulation**
 - B. inheritance
 - C. pointers
 - D. polymorphism
-

Q.7 What will be the output of the following code segment?

```
setprecision (2) << pi ;
```

```
float pi=32.3996;
```

```
A. 32.310
```

```
cout<<fixed;
```

```
B. 32.39
```

```
cout<<32.40
```

```
D. 32.00
```

Q.8 What is the sequence of event(s) when deallocating memory of an object using delete operator?

A. Only block of memory is deallocated for object

B. Only destructor is called for object

C. Memory is deallocated first before calling destructor

D. Destructor is called first before deallocating memory

Q.9 The new operator automatically determines the size of memory required to store that object , so it does not need _____.

A. dot operator

B. free operator

C. size off operator

D. sizeof operator

Q.10 The members of a class declared without any keyword are _____ by default

A. private

B. public

C. constant

D. protected

Q.1 What will be the output of following statement?

```
cout<<setw(10)(<0'>) << setw(7) << 128 ;
```

A. 0128128

C. 0000128

D. 0012800

Q.2 C++ offers _____ levels of data access control inside a class

A. two

B. three

C. five

D. four

Q.3 The ws manipulator can be used with ____ stream(s).

A. None of the given options

B. Input

C. Input/Output

D. Output

Q.4 What will be the output of the following code segment?

```
setprecision (2) << pi ;
```

```
float pi=32.3996;
```

A. 32.39

```
cout<<fixed;
```

B. 32.40

```
cout<<32.00
```

D. 32.310

Q.5 The members of a class declared without any keyword are _____ by default

A. protected

B. private

C. constant

D. public

Q.6 When operator function is implemented as member function then return type of function _____.

A. Must be an object of same class

B. Must be user-defined data type

C. Can be any data type

D. Must be built-in data type

Q.7 If we do not write our own assignment operator then which of the following problem may occur?

- A. Dangling pointer
 - B. Unreferenced memory
 - C. NULL pointer
 - D. Memory leak**
-

Q.8 Which of the following syntax is best used to delete an array of 5 objects named 'string' allocated using new operator.

- A. delete string ;
 - B. delete []string ;**
 - C. delete string[] ;
 - D. delete string[5] ;
-

Q.9 In the scenario, if we want to grant access to the private data members of a Class1 to the Class2, then which statement is true about this scenario?

- A. Class2 is required to be friend of Class1**
 - B. Class1 is required to be friend of Class2
 - C. Class1 inherits from Class2
 - D. Class2 inherits from Class1
-

**Q.10 If text is a pointer of type String then what will be the functionality of following statement?
String [5];**

- A. Creates array of 5 objects statically
 - text = new**
B. Creates array of pointers to string
 - C. Creates array of 5 objects dynamically**
 - D. Creates a string Object
-

Q.1 A friend function of a class is a function defined _____.

- A. outside that class and that has the right to access public members of the class only
- B. outside that class and that has the right to access protected members of the class only.
- C. outside that class and that has the right to access all members of the class.**
- D. inside that class and that has no access to the members of the class

Q.2 What is meant by the following statement?.

String arr[5] = {String("Programming"), String("CS201")};

- A. String constructor will be called for first 2 objects and default constructor for remaining objects
- B. Default constructor will be called for all objects of array
- C. Default constructor will call for first 2 objects and parameterized constructor for remaining objects
- D. Parameterized constructor will be called for all objects of array

Q.3 new operator can be used for _____.

- A. integer , float, char and double data types**
- B. dot operator
- C. only char and integer data types
- D. only integer data type

Q.4 If we have a program that writes the output data(numbers) to the disc ,and if we collect the output data and write it on the disc in one write operation instead of writing the numbers one by one. situation the area where we will gather the number is called

In the above

- A. Stack
- B. Cache
- C. Heap
- D. Buffer**

Q.5 The friend functions are _____.

- A. inline functions
- B. member of a class
- C. void functions
- D. not member of a class**

Q.6 What is the function of the following statement to delete an array of 5 objects named 'arr' allocated using new operator?

- A. Results into syntax error
- B. Deletes all the objects of array
- C. Do not delete any object
- D. Deletes only one object of array**

delete arr;

Q.7 The friend function of a class can have access _____.

- A. to the public data members only
 - B. to the private data members
 - C. to the protected data members
 - D. to public, private and protected data members**
-

Q.8 What happens if new fails to allocate memory in C++?

- A. The program crashes
 - B. It returns NULL
 - C. It returns uninitialized memory
 - D. It throws an exception**
-

Q.9 The members of a class declared without any keyword are _____ by default

- A. protected
 - B. public**
 - C. private
 - D. constant
-

Q.10 The only operator that the compiler overloads for user define data type by default is

- A. Plus (+) operator
 - B. Equal (==) operator
 - C. Assignment (=) operator**
 - D. Minus (-) operator
-

Q.1 A _____ function of a class is defined outside that class scope, but it can access all private and protected members of the class.

- A. Private
- B. Global
- C. Protected
- D. Friend**

Q.2 Friend function declaration can go _____ the class

- A. only outside
- B. in the member function of
- C. anywhere in**
- D. only within

Q.3 A friend function of a class is a function defined _____.

- A. outside that class and that has the right to access public members of the class only
- B. outside that class and that has the right to access protected members of the class only.
- C. outside that class and that has the right to access all members of the class.**
- D. inside that class and that has no access to the members of the class

Q.4 Overloaded member operator function is always called by _____

- A. Class
- B. Compiler**
- C. Object
- D. Primitive data type

Q.5 What will be the output of following statement?

`cout << setw(10) << setfill('0') << setw(7) << 128 ;`

- A. 128000
- B. 0128128
- C. 0000128**
- D. 0012800

**Q.6 If text is a pointer of type String then what will be the functionality of following statement?
String [5];**

- A. Creates array of 5 objects statically
- text = new
B. Creates array of 5 objects dynamically**
- C. Creates array of pointers to string
- D. Creates a string Object

Q.7 `i+=2` is equivalent to _____.

- A. `i=i++;`
- B. `i=1+I;`
- C. `i=i+2;`
- D. `2+=I;`

Q.8 free function is available in _____ header file.

- A. `string.h`
- B. `stdlib.h`
- C. `conio.h`
- D. `iostream.h`

Q.9 What happens when `new` is called to create an object?

- A. The object is immediately destroyed.
- B. Only memory is allocated (no constructor call).
- C. The destructor is called automatically.
- D. Memory is allocated, and the constructor is called

Q.10 What will be the output of the following code segment?

- `int n=8;`
`cout<<oct<<n;`
- A. 8
 - B. 1000
 - C. 11
 - D. 10

Q.1 If we want to allocate memory to an array of 5 integers dynamically, the syntax will be _____.

- A. `iptr= new[5]`
- B. **`int *iptr ;`**
- C. `int iptr ;`
`iptr = new int[5] ;`
- D. `integer iptr** ;`
`iptr= int [5]`

Q.2 Suppose we have a data member `buf` of our `String` class and we like to refer it in a member function of the same class, then choose which option is incorrect?

- A. `this -> buf`
- B. `(*this).buf`
- C. **`this.(*buf)`**
- D. `buf`

Q.3 What is the sequence of event(s) when deallocating memory of an object using delete operator?

- A. Memory is deallocated first before calling destructor
- B. Only block of memory is deallocated for object
- C. **Destructor is called first before deallocating memory**
- D. Only destructor is called for object

Q.4 `int &i;` It means that `i` is a _____ to an integer.

- A. data type
- B. **reference**
- C. pointer
- D. constant pointer

Q.5 _____ data isn't accessible by non-member functions or outside classes

- A. Public
- B. **private**
- C. Static
- D. Globally declared

Q.6 If we have a program that writes the output data(numbers) to the disc ,and if we collect the output data and write it on the disc in one write operation instead of writing the numbers one by one. situation the area where we will gather the number is called

In the above

- A. Cache
- B. **Buffer**
- C. Heap
- D. Stack

Q.7 The operator to free the allocated memory using new operator is _____.

- A. del
 - B. free
 - C. delete**
 - D. remove
-

Q.8 Operator overloading is to allow the same operator to be bound to more than one implementation, depending on the types of the _____.

- A. Applications
 - B. Function names
 - C. Compilers
 - D. Operands**
-

Q.9 Once the _____ are created, they exist for the life time of the program

- A. non static variables
 - B. global variables**
 - C. local variables
 - D. automatic variables
-

Q.10 Memory allocated from heap or free store _____.

- A. cannot be returned back unless freed explicitly using malloc and realloc
 - B. can be returned back to the system automatically
 - C. can be allocated to classes only
 - D. cannot be returned back unless freed explicitly using free and delete operators**
-

Q.1 The friend functions are _____.

- A. member of a class
- B. inline functions
- C. not member of a class**
- D. void functions

Q.2 Overloaded delete operator function takes parameter of void pointer and returns.

- A. int pointer
- B. void pointer
- C. object pointer
- D. void (nothing)**

Q.3 `i+=2` is equivalent to _____.

- A. `2+=I;`
- B. `i=i+2;`**
- C. `i=1+I;`
- D. `i=i++;`

Q.4 If we want to allocate memory to an array of 5 integers dynamically, the syntax will be _____.

- A. `integer iptr** ;`
- B. `int iptr ;
iptr= new int[5]`
- C. `iptr= new[5]
iptr= int [5]`
- D. `int *iptr ;`**

`iptr = new int[5] ;`
Q.5 What is meaning of bad bit in I/O operation:

- A. Wrong data
- B. Lost data
- C. Error in data**
- D. Bad data

Q.6 The members of a class declared without any keyword are _____ by default

- A. private**
- B. protected
- C. public
- D. constant

Q.7 For the joining of two strings in string class, we may use "+" operator, can we use "-" operator the same way for subtracting strings?

- A. Yes along "+" operator only
- B. Yes
- C. Yes along ":" operator only

D. No

Q.8 What will be the output of the following code segment?

```
setprecision (2) << pi ;
```

```
float pi=32.3996;
```

```
cout<<fixed;
```

```
cout<<32.40
```

D. 32.39

Q.9 The concept of _____ allows us to separate the interface from the implementation of the class.

A. encapsulation

- B. pointers
 - C. polymorphism
 - D. inheritance
-

Q.10 A _____ function of a class is defined outside that class scope, but it can access all private and protected members of the class.

- A. Global
 - B. Friend**
 - C. Private
 - D. Protected
-

Q.1 With user data type variables (Objects) self assignment can produce

- A. Logical error
 - B. Run time error**
 - C. Link error
 - D. Syntax error
-

Q.2 What happens if new fails to allocate memory in C++?

- A. It throws an exception**
 - B. The program crashes
 - C. It returns NULL
 - D. It returns uninitialized memory
-

Q.3 _____ data isn't accessible by non-member functions or outside classes

- A. Globally declared
 - B. private**
 - C. Public
 - D. Static
-

Q.4 dec, hex, oct are all _____

- A. Non-parameterized manipulators
 - B. Member functions
 - C. Parameterized manipulators**
 - D. Objects of input/output streams
-

Q.5 Where should delete typically be called to deallocate memory allocated in a constructor?

- A. Immediately after new.
 - B. In the main program.
 - C. In another member function.
 - D. In the destructor**
-

Q.6 What will be the output of the following code segment?

```
setprecision (2) << pi ;  
float pi=32.3996;
```

- A. 32.39**
 - ~~B. 32.310~~
 - ~~C. 32.00~~
 - D. 32.40
-

Q.7 new operator can be used for _____.

- A. dot operator
 - B. only char and integer data types
 - C. only integer data type
 - D. integer , float, char and double data types**
-

Q.8 When we used eof (end of file) with the file reading than which of the following is a true statement?

- A. This is way to check state of the stream**
 - B. This is way to check type of the stream
 - C. This is way to check source of the stream
 - D. This is way to check destination of the stream
-

Q.9 For the joining of two strings in string class, we may use "+" operator, can we use "-" operator the same way for subtracting strings?

- A. Yes along ":" operator only
 - B. Yes along "+" operator only
 - C. Yes
 - D. No**
-

Q.10 Which one of the following is the declaration of overloaded pre-increment operator implemented as member function?

- A. Class-name operator ++();**
 - B. Class-name operator ++(int) ;
 - C. Class-name operator +(int) ;
 - D. Class-name operator +(int) ;
-

Q.1 The friend keyword provides access _____.

- A. to the data members of the friend class only
- B. in two directions
- C. in one direction only**
- D. to all classes

Q.2 The new operator automatically determines the size of memory required to store that object , so it does not need _____.

- A. dot operator
- B. free operator
- C. sizeof operator**
- D. size off operator

Q.3 The friend function of a class can have access _____.

- A. to the public data members only
- B. to public, private and protected data members**
- C. to the private data members
- D. to the protected data members

Q.4 int &i; It means that i is a _____ to an integer.

- A. pointer
- B. constant pointer
- C. reference**
- D. data type

Q.5 Functions declared with the _____ specifier in a class member list are called of that class.

friend functions

- A. private
- B. public
- C. protected
- D. friend**

Q.6 The statement cin.get (); is used to_____.

- A. Read a character from the keyboard**
- B. Read a string from keyboard
- C. Read a String from file
- D. Read a character from file

Q.7 What is meant by the following statement?.

String s[5] = {String("Programming"), String("CS201")};

B. Parameterized constructor will call for first 2 objects and default constructor for remaining objects

- C. Default constructor will be called for all objects of array
- D. Parameterized constructor will be called for all objects of array

Q.8 What is the sequence of event(s) when allocating memory to an object using new operator?

A. Memory is allocated first before calling constructor

- B. Only constructor is called for object
- C. Only block of memory is allocated for object
- D. Constructor is called first before allocating memory

Q.9 The members of a class declared with the keyword struct are _____ by default.

- A. static
- B. private
- C. public**
- D. protected

Q.10 The concept of friend function negates the concept of _____.

- A. inheritance
- B. encapsulation**
- C. polymorphism
- D. persistence

Q.1 The new operator automatically determines the size of memory required to store that object , so it does not need _____.

- A. size off operator
- B. free operator
- C. dot operator
- D. sizeof operator**

Q.2 *this is a pointer that always points to

- A. Current pointer of the class
- B. Current object of the class**
- C. A Value of data member
- D. Current function of the class

Q.3 If the request of new operator is not fulfilled due to insufficient memory in the heap _____.

- A. the new operator returns 2
- B. free operator returns nothing
- C. the new operator returns 1
- D. the operator returns 0**

Q.4 _____will return the number of bytes reserved for a variable or data type.

- A. new operator
- B. sizeof operator**
- C. void pointer
- D. free operator

Q.5 Which statement is true about the following code in C++? int &j = i;

- A. The value of variable i is assigned to the variable j
- B. The address of variable i is assigned to the variable j
- C. j is a synonym of variable i**
- D. The address of variable j is assigned to the variable i

Q.6 Overloaded assignment operator must be

- A. Member function of class**
- B. Global function
- C. Non-member function of class
- D. Friend function of class

Q.7 Memory allocated from heap or free store _____.

- A. can be allocated to classes only
 - B. cannot be returned back unless freed explicitly using free and delete operators**
 - C. can be returned back to the system automatically
 - D. cannot be returned back unless freed explicitly using malloc and realloc
-

Q.8 What will be the output of following statement?

`cout << setfill('0') << setw(7) << 128 ;`

- B. 0000128**
 - C. 0012800
 - D. 1280000
-

Q.9 free function is available in _____ header file.

- A. iostream.h
 - B. conio.h
 - C. string.h
 - D. stdlib.h**
-

Q.10 When the compiler overloads the assignment (=) operator by default then

- A. Class members are not assigned properly
 - B. Compiler does not do member wise assignment.
 - C. Compiler does member wise assignment.**
 - D. Compiler does not allow default assignment operator
-

Q.1 Once an object is declared as a friend, _____.

- A. it has access to all non-public members as if they were public**
- B. it has access to protected data members only
- C. it has access to public members only
- D. it has no access to data members of the class

Q.2 Friend function declaration can go _____ the class

- A. in the member function of
- B. only within
- C. anywhere in**
- D. only outside

Q.3 $i+=2$ is equivalent to _____.

- A. $2+=I;$
- B. $i=i++;$
- C. $i=1+I;$
- D. $i=i+2;$**

Q.4 When operator function is implemented as member function then return type of function _____.

- A. Must be built-in data type
- B. Can be any data type**
- C. Must be user-defined data type
- D. Must be an object of same class

Q.5 The members of a class declared without any keyword are _____ by default

- A. public
- B. constant
- C. protected
- D. private**

Q.6 When the compiler overloads the assignment (=) operator by default then

- A. Compiler does member wise assignment.**
- B. Compiler does not allow default assignment operator
- C. Compiler does not do member wise assignment.
- D. Class members are not assigned properly

Q.7 If we use cin stream to read some value and store it in some integer variable and press some alphabet key instead of numeric keys. then what will happen?

- A. Its binary representation will be ignored and the value will be stored
 - B. Its ASCII code will be stored inside the computer**
 - C. Its binary representation will be ignored and the character will be stored
 - D. Some error will occur and cin stream will detect this error.
-

Q.8 Which of the following is a destination of cout stream?

- A. Constant
 - B. Variable
 - C. Printer**
 - D. Data type
-

Q.9 If operator function is non-member function then object on left side of operator cannot be_____?

- A. Built-in data type
 - B. Object of different class
 - C. Object of member function
 - D. Object of operator class**
-

Q.10 *this is a pointer that always points to

- A. A Value of data member
 - B. Current pointer of the class
 - C. Current object of the class**
 - D. Current function of the class
-

Q.1 If we do not write our own assignment operator then which of the following problem may occur?

- A. Unreferenced memory
 - B. Dangling pointer
 - C. NULL pointer
 - D. Memory leak**
-

Q.2 The operator to free the allocated memory using new operator is _____.

- A. delete**
 - B. free
 - C. del
 - D. remove
-

Q.3 Which of the following functionality can be achieved through overloading?

- A. We can change the precedence of some operators.
 - B. We can change the associativity of some operators.
 - C. New operators cannot be defined through operator overloading.**
 - D. We can change associativity of all operators.
-

Q.4 If the request of new operator is not fulfilled due to insufficient memory in the heap _____.

- A. the new operator returns 2
 - B. free operator returns nothing
 - C. the operator returns 0**
 - D. the new operator returns 1
-

Q.5 The members of a class declared with the keyword struct are _____ by default.

- A. static
 - B. private
 - C. protected
 - D. public**
-

Q.6 Friend classes are used in cases where one class is _____ to another class

- A. encapsulated
 - B. loosely coupled
 - C. independent
 - D. tightly coupled**
-

Q.7 Which of the following is normally a destination for cin stream?

- A. Variable
 - B. Keyboard**
 - C. Mouse
 - D. Printer
-

Q.8 _____ must be included to use stream manipulator in your code.

- A. stdlib.h
 - B. iostream
 - C. iomanip**
 - D. conio.h
-

Q.9 int &i; It means that i is a _____ to an integer.

- A. reference**
 - B. constant pointer
 - C. data type
 - D. pointer
-

Q.10 In overloading the assignment (=) operator, which object(s) will call the operator function?

- A. Right object of the assignment operator
 - B. No object will call the operator function
 - C. Left object of the assignment operator**
 - D. Both objects will call the operator function
-

Q.1 What is the sequence of event(s) when allocating memory to an object using new operator?

- A. Only block of memory is allocated for object
 - B. Memory is allocated first before calling constructor**
 - C. Constructor is called first before allocating memory
 - D. Only constructor is called for object
-

Q.2 The friend function of a class can have access _____.

- A. to the protected data members
 - B. to public, private and protected data members**
 - C. to the private data members
 - D. to the public data members only
-

Q.3 For the joining of two strings in string class, we may use "+" operator, can we use "-" operator the same way for subtracting strings?

- A. Yes along ":" operator only
 - B. Yes along "+" operator only
 - C. No**
 - D. Yes
-

Q.4 Which one of the following is the declaration of overloaded pre-increment operator implemented as member function?

- A. Class-name operator ++();**
 - B. Class-name operator +(int) ;
 - C. Class-name operator ++(int) ;
 - D. Class-name operator +() ;
-

Q.5 Which of the following is a destination of cout stream?

- A. Data type
 - B. Constant
 - C. Variable
 - D. Printer**
-

Q.6 Suppose we have a data member buf of our String class and we like to refer it in a member function of the same class, then choose which option is incorrect?

- A. (*this).buf
 - B. this.(*buf)**
 - C. this -> buf
 - D. buf
-

Q.7 Memory allocated from heap or free store _____.

A. cannot be returned back unless freed explicitly using free and delete operators

B. can be allocated to classes only

C. can be returned back to the system automatically

D. cannot be returned back unless freed explicitly using malloc and realloc

Q.8 What is the purpose of endl stream manipulator?

A. terminate a string with NULL

B. flush the stream

C. insert a new line

D. insert a new line and flush the stream

Q.9 $i+=2$ is equivalent to _____.

A. $2+=i$;

B. $i=1+i$;

C. $i=i++$;

D. $i=i+2$;

Q.10 Choose the operator which cannot be overloaded in C++.

A. ?

B. ~

C. | |

D. &
